A Project Report
on
ACEMS

Submitted by

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Under the Guidance of
Mr. Raunak Raj

In partial fulfillment for the award of the degree of

BACHELOR OF TECHNOLOGY
in
INFORMATION TECHNOLOGY

PARUL INSTITUTE OF ENGINEERING AND TECHNOLOGY,
PARUL UNIVERSITY,
VADODARA, GUJARAT

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CERTIFICATE

This is to certify that the Project Report entitled, "**ACEMS**" submitted by "**MAJMUNDAR HARSH, THAKOR HARSHRAJ, PATEL DHRUV, PARMAR DIVYASINH"** to **Parul University, Vadodara, Gujarat**, is a record of Bonafide Project work carried out by them under my supervision and guidance, and is worthy of consideration for the award of the degree of **Bachelor of Technology** in **Information Technology** of the University.

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ACKNOWLEDGEMENT

Behind any major work undertaken by an individual there lies the contribution of the people who helped them to cross all the hurdles to achieve their goal.

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We also express our deep sense of regards and thanks to Dr. Pooja Sapra (Professor), Head of INFORMATION TECHNOLOGY Engineering Department. We feel very privileged to have her precious advices, guidance and leadership.

ABSRACT

The influencer marketing industry frequently faces challenges which include contract disputes and delayed payments together with disputes about deliverables and the lack of neutral enforcement methods which should exist between brands and content creators. The project introduces The Autonomous Contract Execution and Monitoring System (ACEMS) which operates as a backend third-party system that automatically manages contract processes through unbiased and secure methods. The system achieves contract execution through its automated processes while creating equitable conditions for all aspects of contract execution.

ACEMS system operates through secure API layers which enable users to manage campaigns and contracts while using an AI-powered creator–brand matching system that matches creators with brands according to their performance metrics. The system includes a rule-based state machine which enforces contract rules and maintains contract terms until both parties reach mutual agreement. The system uses a monitoring module which provides real-time updates on deliverable and deadline progress. The system achieves contract execution through automated compliance checks which provide transparent contract records during all stages of the contract lifecycle. The system maintains consistent contract execution through its combination of various operational elements.

The system improves operational fairness through its elimination of human decision-making which executes compliance checks and payment processing. The system demonstrates both modular operation and scalable design through its current capabilities which will develop advanced contract analysis together with cross- platform system connectivity in upcoming times. The system provides a trustworthy base which supports autonomous contract management through digital creator economy platforms while maintaining transparent and impartial contract management operations.

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CHAPTER 1 INTRODUCTION

1.1. PROBLEM STATEMENT:-

Influencer marketing industry has expanded very fast, yet it is still struggling with serious operation issues concerning contract execution and compliance management. The brands and content creators are prone to conflict because of subjective explanations of the terms of delivery, late payments, and the lack of neutral enforcement tools. The current platforms are very manual-driven, creating bias, fluctuations, and ineffective processes during the entire contract lifecycle.

The existing influencer marketing systems do not have automatic systems to guarantee the immutability of contracts once they are made. This permits post-approval changes, irregular decisions on compliance and disagreements after the campaign has been executed. Moreover, the payment disbursement is in many cases delayed because of the need to manually check the deliverables, which causes the lack of trust between the stakeholders and low transparency in the contractual performance.

The second significant shortcoming of the current workflows is that there is no real-time compliance monitoring and objective decision-making. The majority of platforms rely on human man to analyze the fulfillment of contractual obligations, and it may lead to inaccuracy, biased interpretation, or bias. It is also not very much provided to keep auditable records that effectively explain compliance decisions and payment results.

Thus, it is highly desired that a secure and automated and unbiased system can handle influencer contracts end-to-end, without human influence. The issue to this project is the deficit of a scalable backing framework that has the ability to implement contract logic and track deliverables in real-time and perform payments in a transparent manner. The suggested Autonomous Contract Execution and Monitoring System (ACEMS) seeks to resolve these issues by giving an immutable, rule-based, and auditable contract execution system to influencer marketing systems

1.2 MOTIVATION:-

The accelerated rise of digital platforms and social media has made influencer marketing highly important in the contemporary advertising practices. The use of influencers to advertise products and services is on the rise among brands owing to their capacity to connect with the right consumers and create trust. Nevertheless, the trade procedures of influencer collaborations are quite manual and poorly controlled even though they are very popular and economical. This poses some difficulties in contract management, content deliverability, and payment between content creators and brands on time.

Among the key incentives to drive this project, we can recognize the number of conflicts, which are often arising because of the ambiguity of terms of a contract and the possibility of subjective understanding of campaign deliverables. Influencer contracts usually consist of several clauses connected to the posting schedule, the quality of content, the level of engagement, and the promotion. In case these conditions are kept by hand, it is hard to keep consistency and fairness in the decision-making process. Consequently, there is often a conflict among the stakeholders on whether they have met the terms and conditions of the contract, and this matters in the end to the ecosystem trust.

The majority of current influencer marketing solutions rely on the internal departments or the platform managers to assess compliance and send money. This may lead to bias, inconsistency and delays in decision-making, as such reliance on human judgment is possible. Moreover, due to the lack of formal monitoring mechanisms, it is hard to keep credible records that can be used to support compliance results. Such transparency lack leads to less accountability and greater chances of conflict once the campaigns are over.

The rationale of the Autonomous Contract Execution and Monitoring System (ACEMS) creation is thus to overcome such hurdles with the introduction of an automated and transparent contract management system without being biased in influencer contract management. The system will enhance the level of trust between brands and creators since the execution of the contract will be carried out fairly and consistently. Removing the subjective side of decision-making and providing an opportunity to monitor campaign deliverables in real-time, ACEMS plans to minimize any controversy, enhance efficiency in operations, and create a stable governance system within the digital creator partnerships.

1.3 Aim & Objective:-

The main goal of the following project is to have an automated backend system that will be able to handle the lifecycle of influencer marketing contracts in a secure, transparent, and unbiased way. It is the system that seeks to remove manual access to monitoring and executing contracts and implement rule-based contract enforcement systems with automatic compliance checks.

The other objective is to make sure that the agreement is immutable as soon as both the parties agree with their contracts. The system ensures that no unauthorized changes are made to the contracts once approved since it does this by locking the contracts such that all the stakeholders strictly follow the terms and conditions agreed upon. This assists in ensuring equitability and consistency in agreement of the contract and minimizes chances of conflict.

Another goal is the creation of a real-time monitoring system that is able to track campaign deliverables and deadlines. The system must have the automated check of the fulfillment of contractual obligations and give the relevant actions like payment disbursement or penalty application. This will guarantee the right time implementation and enhance accountability in the influencer marketing processes.

Lastly, the project will also develop a scalable and modular system architecture that can be extended through future developments to include AI-driven contract analysis, advanced compliance assessment, and connection to various social media platforms. By means of these goals, the Autonomous Contract Execution and Monitoring System (ACEMS) proposed is aimed at offering a trusted framework under which the management of the influencer marketing contracts may be carried out effectively and without bias.

1.4 SCOPE:-

This project scope is dedicated to the design and development of an automated contract execution and monitoring system in the influence marketing platforms based on the backend. The system concerns mainly the difficulties associated with enforcement of the contract and verification of compliance and execution of payment between the brands and creators of the content. ACEMS is supposed to be a neutral back office entity, which makes the contract lifecycle fair and consistent.

The project encompasses the initiation of secure contract management capabilities, such as creation of contracts, contract approval, immutability, and rule-based execution of contracts. It also incorporates the real-time tracking of the campaign deliverables and deadlines to ensure the compliance is achieved automatically. The system can operate in areas such as automatic disbursement of payments and imposition of penalties according to set contract policies.

The area is also expanded to the implementation of an AI-based creator-brand matching platform that assists in the determination of appropriate influencers based on the specifics of the campaign and the performance indicators of the past. The system also has systematic and open audit trails that document all the contract activities and hence accountability and traceability of decisions. Nevertheless, the project is at the present restricted in the general aspects of the development of the backend system, which does not involve frontend user interface development and direct connectivity to external social media. The further development of such functions as drafting legal contracts, resolving disagreements with the help of the legal systems, and real-time analytics in social media is regarded as out of scope of the current implementation.

As regards the future scope, the system can be scaled to accommodate superior AI-based contract analysis, cross-platform integrations, and better compliance assessment procedures. ACEMS is scalable and adaptable with the modular architecture, which is why it can be adopted into various digital creator ecosystem

CHAPTER 2 LITERATURE REVIEW

2.1 Critical Evaluation of General Papers:-

Sr. No.	Publication Year	Author	Publication House	Summary	Research Gaps
1	2024	Hao Ding, Yizhou Liu, Xuefeng Piao	SSRN	The system detects vulnerabilities in smart contracts through its combination of CodeBERT-based RAG and Self-Check Chain-of-Thought which achieves better results than conventional static analysis tools.	The approach breaks down when the prompt gets too long, and if the base model hallucinates on complex logic, the self-check mechanism has no real fallback to catch it.
2	2025	P.S.G. Arunasri, Phani Kumar Solleti, M. Sailaja, P. Mathiyalaga n, Kathiravan G.K., M. Soma Sabitha	SCITEPRESS	The system takes contract elements from legal documents and transforms them into blockchain modular templates which use Solidity and Chaincode to create dynamic contract clauses. The system tests these templates through real-world simulations as a method of validation.	Everything was tested on simulations only with no real-world deployment evidence. Mapping legal nuances cleanly across different jurisdictions remains a hard, unsolved problem.
3	2024	Moez Krichen, Mariam Lahami, Qasem Abu Al-Haija	Journal of Network and Computer Applications	A comprehensive literature review on theorem proving, model checking, and software testing approaches to verify smart contracts.	These techniques demand specialist knowledge and don't scale well to large contracts. The tools themselves aren't mature enough for widespread practical adoption yet.

4	2025	Tudor Ferariu,Philip Wadler, Orestis Melkonian	FMBC 2025	The research presents a comprehensive examination of current formal methods used to create smart contract specifications, which the authors organized into different categories based on their respective modeling and testing capabilities.	The paper is a review only with no new tool or fresh evaluation. Some referenced work may already be outdated by 2026, and no performance comparisons are provided between approaches.
5	2025	Alfred Kuhlman, Arya Wicaksana	JOWUA	Refining MiscSkill-Inquire (7B) in a productive exercise. These units were to be enrolled to generate Contract data, in reference to a specific Gas.(one model) .	Output quality depends entirely on the fine-tuning dataset. The model can still introduce subtle logical errors, and it was only tested on a limited set of contract scenarios.
6	2025	Joongho Ahn, Moonsoo Kim	Electronics	These seven- layerally manned Al agents-what might be in traditional computer programing parlance termed "persona"-to operate on ElizaOternity personalization personas on social media functionalities.	Good user ratings don't guarantee real adoption. Performance was inconsistent with noticeable latency differences across platforms, and scalability remains unproven.

7	2024	Ye Liu, Yue Xue, Daoyuan Wu, Yuqiang Sun, Yi Li, Miaolei Shi, Yang Liu	NDSS 2025	The system uses RAG together with GPT-4 for automatic creation of formal verification properties which results in 80% recall performance and detection of 26 already established vulnerabilities plus 12 previously unknown security risks.	The pipeline depends heavily on the quality of the reference property database. The LLM can still generate incomplete properties, and an external prover is always required.
8	2023	Tharaka Mawanane Hewa, Yining Hu, Madhusanka	IEEE	The survey provides a complete analysis of smart contract platforms through its evaluation of technical attributes and its identification of existing challenges and upcoming research areas.	Transaction throughput and latency are still bottlenecks. Critical coding vulnerabilities keep appearing, and no unified solution has emerged to address them across different platforms.
9	2024	Reto Hofstetter, Andreas Lanz, Navdeep S. Sahni	Journal of Industrial Information Integration	The research analyzed approximately 1000 real influencer contracts which showed that relaxing contractual limitations resulted in three times higher influencer retention rates while maintaining the same financial expenses.	The study covers only traditional non- blockchain contracts. Findings are specific to the observed platforms, and no automated execution or monitoring system is proposed.

10	2023	Bagam Laxmaiah, S. Subburam	IEEE	The solution combines machine learning models with smart contract technology to enable continuous monitoring of fake product detection which results in improved supply chain security.	Regulatory compliance and cross- platform interoperability remain tricky. Scaling the system and managing new risks introduced by full automation haven't been fully addressed.
11	2024	Cristian Valencia-Payan, David Griol, Juan Carlos Corrales	Journal of Logic and Computation	An autonomous smart contract feeding itself with an integrated ML anomaly detection working in a supply chain context at a speed of 184 tps with latency of only 0.41sec.	More work is needed for complex or evolving supply chain disruptions. The ML component lacks detailed documentation, and the system is scoped only to traceability use cases.
12	2025	Saad AL Azzam, Raenu AL Kolandaisa my, Ghassan AL Dharhani	Mesopotamian Journal of Big Data	The study systematically reviews artificial intelligence machine learning and deep learning techniques that researchers have developed to secure smart contracts which concrete audits fail to detect complex logical vulnerabilities.	Interoperability across chains is still a barrier. Legal enforcement lags behind technical capability, and sharing contract data across parties raises unresolved privacy concerns.

13	2024	Sirui Hong, Mingchen Zhuge, Jiaqi Chen	ICLR 2024	The meta- programming framework uses structured prompts to create standard operating procedures which it implements through its role- based multi-agent system in order to decrease false information and enhance its ability to handle tasks.	The framework is only as strong as the underlying LLM. Writing effective SOPs requires significant care, and evaluations have mostly been limited to software engineering tasks.
14	2024	Tomasz Górski	Applied Sciences	The system transforms smart contract source code into graph formats which it uses to develop deep learning models. The system removes the need for manual rule development through its automatic rule generation capability.	Developers must follow a specific object-oriented or functional programming structure, which creates a real barrier for beginners and limits broader adoption of the pattern.

15	2024	Yu Sun, Daoyuan Wu, Yue Xue, Han Liu	IEEE	The system establishes specific scenarios for testing logic vulnerabilities while using static analysis to verify GPT results, which enables fast and affordable detection of advanced security weaknesses.	Reasoning quality is capped by the underlying model's capability. Precision drops noticeably when contracts being analyzed are very large or structurally complex.
16	2021	T. Pujiati, M. Kamil, N. Silawati, RS Ikhsan	ADI Journal	The study used mixed methods to combine big data analysis with case studies and it proved that demand forecasting accuracy and supply chain risk resilience had both achieved substantial improvements.	Findings are tied to specific industry case studies and don't generalize easily. There is also no ethical governance framework guiding AI decisions in high-stakes supply chain contexts.
17	2025	Zhiyuan Wei, Jing Sun, Yuqiang Sun	IEEE	The Double-Mode analysis (.broad target and .adding2) achieves 98% accuracies on the common vulnerabilities, and even with shared auditing costs, contract costs can be dramatically reduced.	The targeted analysis mode is computationally expensive. Both performance and cost scale with contract size, making it impractical for very large or complex codebases.

18	2024	Iqra Mustafa, Alan McGibney, Susan Rea	Frontiers in Blockchain	The GRV-SC engineering framework employs Colored Petri Nets together with a type-safety verifier to provide complete coverage of smart contract development while detecting access control violations.	The framework is tied specifically to the DAML platform. Formal modeling adds meaningful upfront effort, and newly emerging vulnerability types not anticipated at design time can still go undetected.
19	2023	A. Barunaha, MR Prakash, R. Naresh	Atlantis Press	Monitors the Twitter feed, extracting relevant tweets that carry such business significance. According to them, posts with negative propensity may be integral to the analysis, which could then be refreshed into the model through the learning mechanism classic to this kind of supervised learner.	The classifier struggles with sarcasm and irony, two things that show up constantly in social media. Overall accuracy sits at around 76%, which leaves a lot of room for misclassification.
20	2021	Ree Chan Ho, Muslim Amin, Kisang RyuCorrespo nding Author, Faizan Ali	Emerald Publishing Limited (journal of hospitality & tourism technology)	This study develops an "integrative model" based on the UTAUT framework to explain "travelers' continued use of smart travel app itineraries", finding that "hedonic and utilitarian values" strongly influence their intention to use these digital tour plans.	The approach only captures state- machine behavior, so anything more dynamic falls outside its scope. Because deployed contracts are immutable, there's no way to push runtime upgrades once the contract is live.

2.2 SUMMARY OF RESEARCH PAPER:-

- Multiple studies investigate the automatic generation and validation and security of smart contracts through artificial intelligence and formal verification techniques.
- The combination of LLMs with retrieval-augmented generation (RAG) systems leads to better accuracy in smart contract vulnerability detection according to research papers which utilized CodeBERT and PropertyGPT.
- The Code LLaMA model allows users without specialized training to create working and cost-effective Solidity contracts through its specialized programming capabilities.
- Theorem proving and model checking serve as formal verification methods which provide complete correctness proof but their usage faces challenges because of their complicated nature and unavailability of mature tools.
- The MetaGPT multi-agent LLM framework divides complex contractual work into different expert roles which helps to decrease overlapping errors while producing better results.
- The GPTScan and PropertyGPT tools use GPT-based reasoning together with static program analysis to identify both common and zero-day logic vulnerabilities which they handle with high detection rates and minimal expenses.
- The smart contracts that employ self-updating mechanisms together with ML models achieve operational capabilities of 184 tps and 0.41s latency which makes them ideal for real-time supply chain tracking.
- The execution of legal contracts on blockchain technology faces challenges because different jurisdictions require distinct regulations and there may be a need for off- chain legal processes.
- Smart contract life-cycle management frameworks demand complete development process coverage because this method enables early detection of access control.

2.3 LIMITATION/DRAWBACKS OF EXISTING SYSTEM:-

- The first smart contract validation system needs two particular elements because it cannot process extensive contracts during a single validation session.
- Existing vulnerability detection tools require base LLM model quality yet they produce false results when the model encounters intricate logical structures.
- Current formal verification methods require developers to possess advanced expertise because they only support experts through theorem proving and model checking.
- Formal verification tools face major scalability challenges because they cannot process extensive smart contracts which contain multiple nested elements.
- Existing systems support only particular blockchain platforms such as Ethereum and Hyperledger which creates difficulties for users who want to use their systems across different platforms or jurisdictions.
- Smart contracts need off-chain legal systems to resolve their disputes because currently existing solutions do not provide complete automated processes.
- Most AI-based contract generation tools depend heavily on the quality of their fine-tuning dataset and the LLM introduces subtle logical errors which remain hidden.
- Existing multi-agent frameworks like MetaGPT still depend on careful SOP design and have mainly been evaluated on software engineering tasks which limits their ability to function in different situations.
- Current supply chain smart contract systems can only trace specific materials because their systems use limited traceability functions which cannot deal with evolving operational interruptions.
- The interoperability challenges which smart contract vulnerability detection systems experience create major obstacles for their implementation across various blockchain networks and legal systems.
- Privacy concerns related to shared blockchain data remain unresolved because sensitive contract information continues to exist on public and semi-public ledgers.
- Smart contracts executed on blockchain technology face legal enforcement gaps because their operation does not extend legal validity across different jurisdictions.
- The LLM-powered vulnerability detection tools which use GPTScan experience major accuracy reduction when they process smart contract projects which consist of extensive detailed components or sophisticated elements.

CHAPTER 3 PROBLEM DEFINITION AND REQUIREMENT ANALYSIS

3.1 Problem Defination: -

The fast evolution of influencer marketing has revealed a number of operational and trust-related challenges of the existing contract management systems. The majority of platforms depend on manual contract management and human judgments to check deliverables and compliance that usually causes delays, discrepancies and conflicts between brands and content creators. Since there are no standardized mechanisms of enforcement, it is hard to agree that both parties should follow terms of the contract in a fair manner.

The other significant issue is lack of contract impossibility once approved. The contracts in most instances can be changed or reinterpreted once they have been received though that creates misunderstanding and conflicts in the post campaign. Late payments due to slowness in the verification process are another factor that affects less trust and satisfaction amongst creators. Besides, transparency is not very much evident since most of the systems lack adequate audit trails or live monitoring campaign activities.

The above difficulties underscore the necessity of having a safe, computerized, and objective contract execution system. The issue that this project seeks to solve is the absence of an independent mechanism that has the ability to lock contracts once given the green light, constantly surveil deliverables and imposes payments or fines unless the human touch is applied. All these issues need to be addressed with the aim of enhancing fairness, efficiency, and reliability in the management of influencer marketing contract.

3.2 Requirement Analysis : -

The requirement analysis is an important process during system development of ACEMS, since it determines what the system should attain to address the problems identified. The main objective of this step is to know the expectations of the users and translate them into understandable system functions. The system requirements are derived after evaluating the current contract management constraints in order to provide automation, transparency, and equity in the life of the contract.

Brands, content creators, and administrators, with the primary need and access levels, should be supported by the system. Brands need to have an easy system to generate, post, and administer contracts, and creators need straightforward systems to apply to each campaign, deliverables, and payment status. IT administrators need complete access to system monitoring, rule setting, and resolution of disputes.

Functional requirements concentrate on automated contract execution, rule-based compliance check, real-time deliverable tracking and automated payment processing. Non-functional requirements focus on the security, scalability, performance and reliability factors in order to provide a smooth running of the system. Sound requirement analysis will make sure that ACEMS fulfills the technical requirement and the user expectation providing a solid base to the system design and implementation.

3.2.1 User Requirements : -

User requirement is the expectation and requirement of all the stakeholders who interact with the ACEMS platform. The main consumers of the system are the brands, content creators and administrators of the system, who need to have certain functionalities to carry out their duties effectively. These requirements are essential in ensuring that the system is not complicated, is reliable and also that it complies with real life use cases.

The brands need a straightforward and safe application to design campaigns, post contracts, outline deliverables, and provide terms of payment. They should also have real-time access to contract status, compliance by the creator and execution of payments. Content producers are demanding easy access to the listing of the campaign, easy acceptance of contracts with hitches, well-established procedures of submitting the deliverables, and clear tracking of the payment without delays.

Administrators must have full control over the system to track the activities in the platform, regulate the rules, and guarantee system integrity. Every user needs a secure authentication, roles and valid notifications. With these user requirements, the interaction process, trust, and workflow across the contract management system will be smooth.

3.2.2 Functional Requirements :-

- Functional requirements explain the operations and features that the ACEMS system will be required to carry out. These requirements determine how the system is going to behave and how various components will relate to each other to make sure the contract is run and monitored smoothly. They are concerned with automation of processes which are being done manually in the present systems.
- The system should enable the brands to design and enter contracts, specify the campaign guidelines, and choose creators according to the pre-established parameters. It needs to assist in approving contracts automatically, locking approved contracts and implement the rule without human intervention. Creators should also be allowed to apply to campaigns,

accept contracts, and deliverables within given deadlines through the system.

- Also, the system is supposed to monitor the deliverables in real time, verify compliance automatically, and automatically trigger payment or penalties. The functional requirements also include audit logging, notification services, and administration monitoring tools to maintain accountability, transparency and reliability of the system.

3.2.3 Non-Functional Requirements : -

- The non-functional requirements specify quality, the performance, and the reliability requirements that the ACEMS system shall have. These specifications make the system run in an efficient and safe manner, at varying conditions. Although they are not detailed of particular functions, they have a significant influence on user experience and trustworthiness of the system.
- It should also have robust security capabilities, such as secure authentication, encryption of the data and against unauthorized access. It is necessary to have high performance and speed in response time to ensure the users will have access to contract information and a current system response as quickly as possible. The system must be scalable to serve an increasing number of users, campaigns and contracts without compromising the performance.
- The other important one is reliability and availability which means that there would be minimum downtime and constant operation. The system should be able to keep correct audit records and should provide fault recovery. These non-functional requirements will make sure that ACEMS is reliable and secure enough to support real-life workload.

CHAPTER 4 DESIGN AND IMPLEMENTATION

4.1 Design :-

Design phase is the basis of the proposed system and is aimed at transforming the conceptual requirements into a systematic and implementable system model. This step makes sure that system is synchronized to the expectations of the users, functional requirements and long term scalability objectives.

- **Understanding User Needs:**

  Identifying the needs of the stakeholders such as brands, content makers, and platform administrators is the starting point of the design process. The step assists in specifying functionalities of the system without causing unevenness, transparency, and usability.
- **Prototyping the System Architecture:**

  A finer system blueprint is then developed based on the obtained requirements to describe the general sequence of the work and interaction between the modules. This involves defining stages of the contract lifecycle, implementation flows depending on rules, monitoring and data storage forms.
- **Selection of Technology:**

  Practicality in real-time processing, safe transactions, and modular development is the basis of the selection of the databases, API services, and automation tools. The choice of technology stack is made to make sure that it is extensible and can be integrated in the future.
- **Security Factors:**

  The issue of security is also a major concern of the design stage because the system will deal with the sensitive contract and payments information. The design includes measures like access control, data encryption, safe authentication, and role-based permissions. The mechanisms enable integrity of data, avoid unauthorized access and trust among the users of the system.

4.1.1 Use Case Diagram : -

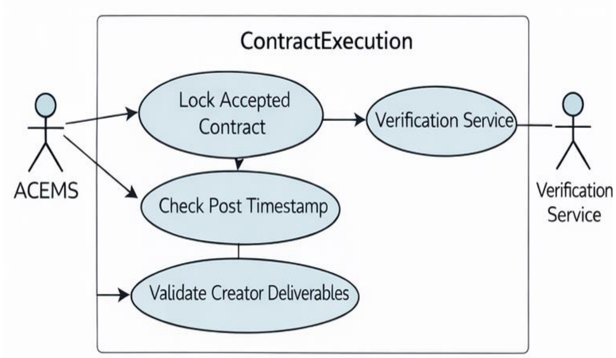

Figure 4.1.1.1 Contract Execution

This above figure represents the Contract Execution process in the ACEMS system. Once a contract is accepted by both parties, ACEMS first locks the contract to prevent any further changes. The system then interacts with an external Verification Service to confirm the contract details. After verification, ACEMS checks the post timestamp to ensure deliverables are submitted on time. Finally, the system validates the creator's deliverables to determine compliance before proceeding with payment or penalties.

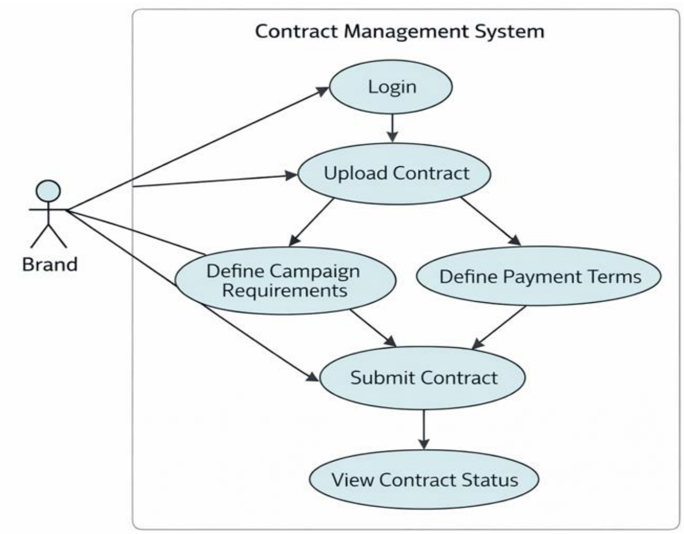

Figure 4.1.1.2 Brand Contract Management System

This diagram shows how a brand interacts with the contract management system. The brand first logs in and uploads a contract related to a campaign. It then defines campaign requirements and payment terms clearly before submitting the contract. After submission, the brand can continuously view the contract status to track approval and progress. This ensures organized and transparent contract creation from the brand's side.

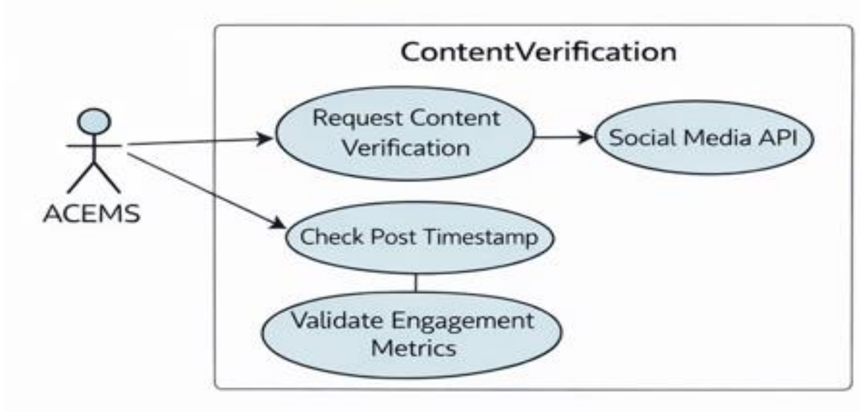

Figure 4.1.1.3 Content Verification System

Here, Content Verification system sends a Request Content Verification in response to the ACEMS actor requesting it. The system then communicates with the Social Media API to retrieve the necessary information on the post. Then, it checks the post date, i.e. whether it was within the required time frame or not. Finally, it improves the confirmation of engagement indicators (likes, comments, shares, etc.) to make sure that the content is of the proper standards of verification.

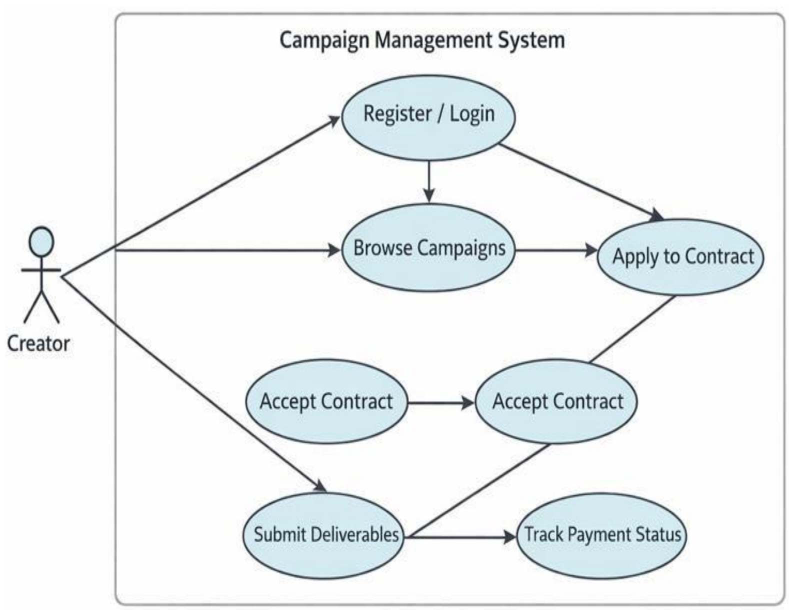

Figure 4.1.1.4Creator Campaign Participation

This diagram explains the process followed by content creators in the system. The creator registers or logs in, browses available campaigns, and applies for a suitable contract. Once selected, the creator accepts the contract and submits the required deliverables. The system also allows creators to track their payment status, ensuring clarity and trust throughout the campaign lifecycle.

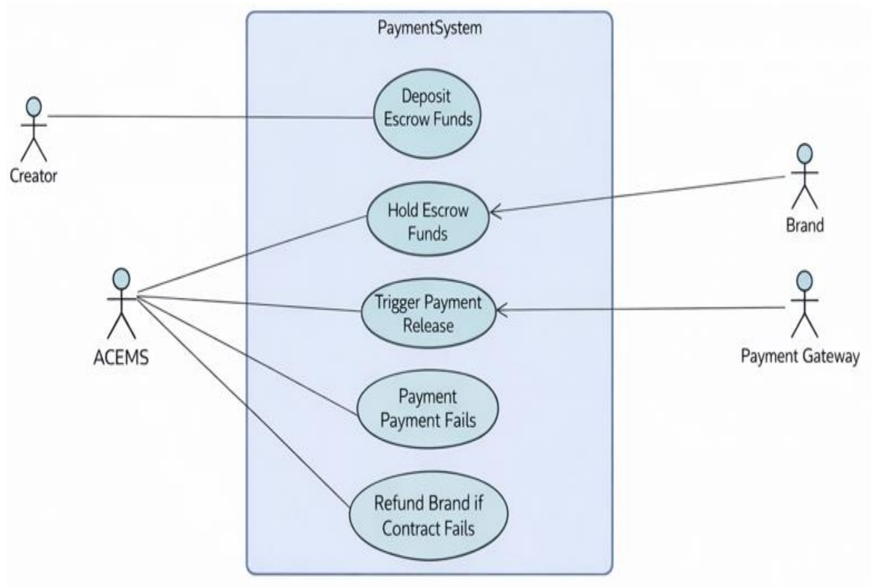

Figure 4.1.1.5 Payment System

This diagram shows how money is handled securely in a payment system using escrow. The brand deposits funds, which the system holds until certain conditions are met. When the contract is successfully completed, the system triggers payment release through the payment gateway to the creator. If the contract fails or payment cannot be processed, the system refunds the brand. It ensures fairness by protecting both sides until rules are satisfied.

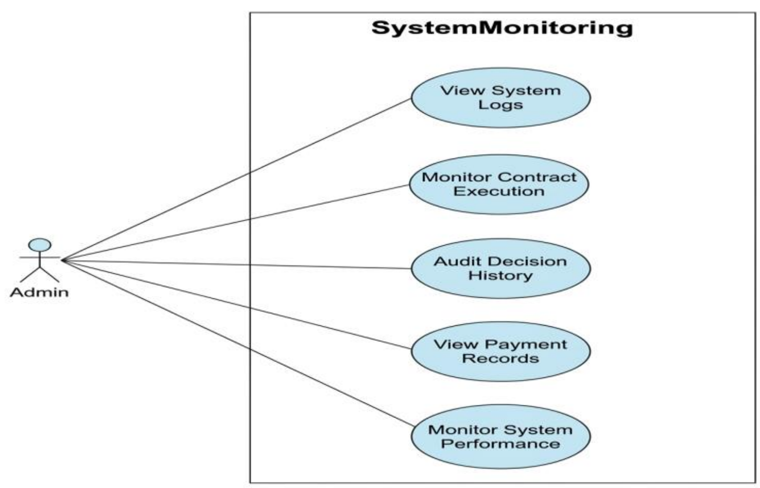

Figure 4.1.1.6 Admin Monitoring System

This diagram highlights what an admin can do to oversee the system. The admin can view logs, monitor contract execution, audit past decisions, check payment records, and track overall system performance. Each function gives visibility into different aspects of the system, helping maintain transparency and reliability. In short, it’s a dashboard of tools for keeping everything running smoothly and accountable**.**

4.1.2 Flow Chart of System : -

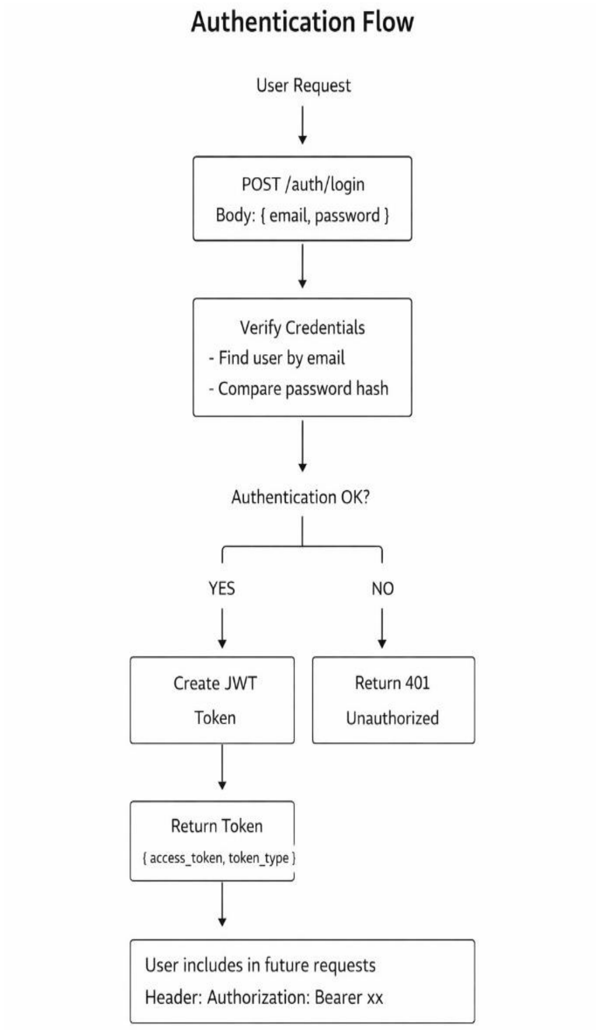

Figure 4.1.2.1 Authentication Workflow Diagram

This diagram shows how a user logs in with email and password. The system checks credentials by finding the user and comparing the password hash. If correct, it creates a JWT token and sends it back. The user then includes this token in future requests to prove they’re authenticated. It’s a secure way to manage login session.

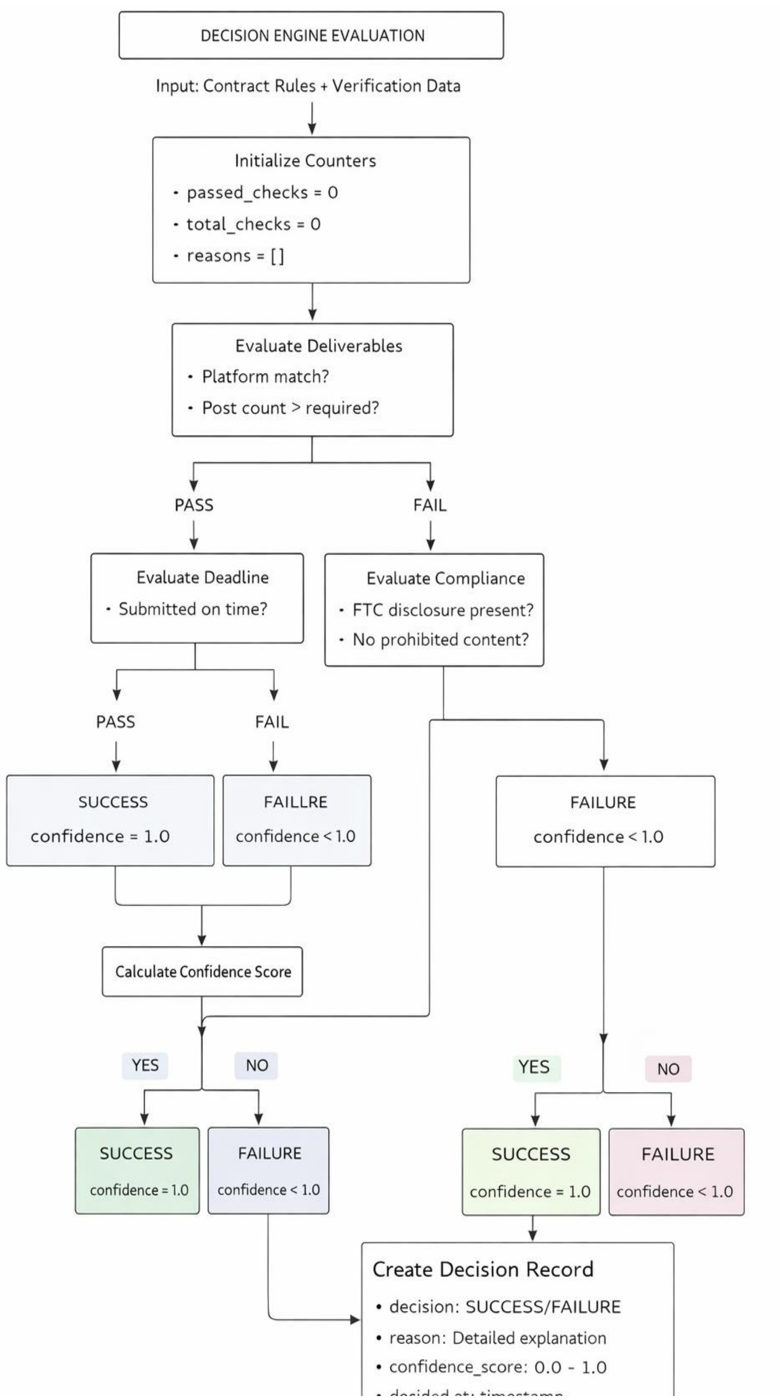

Figure 4.1.2.2 Decision Logic System

This flow explains how the system decides if a contract execution is successful. It checks deliverables (like platform match and post count), deadlines, and compliance rules (such as disclosures and prohibited content).

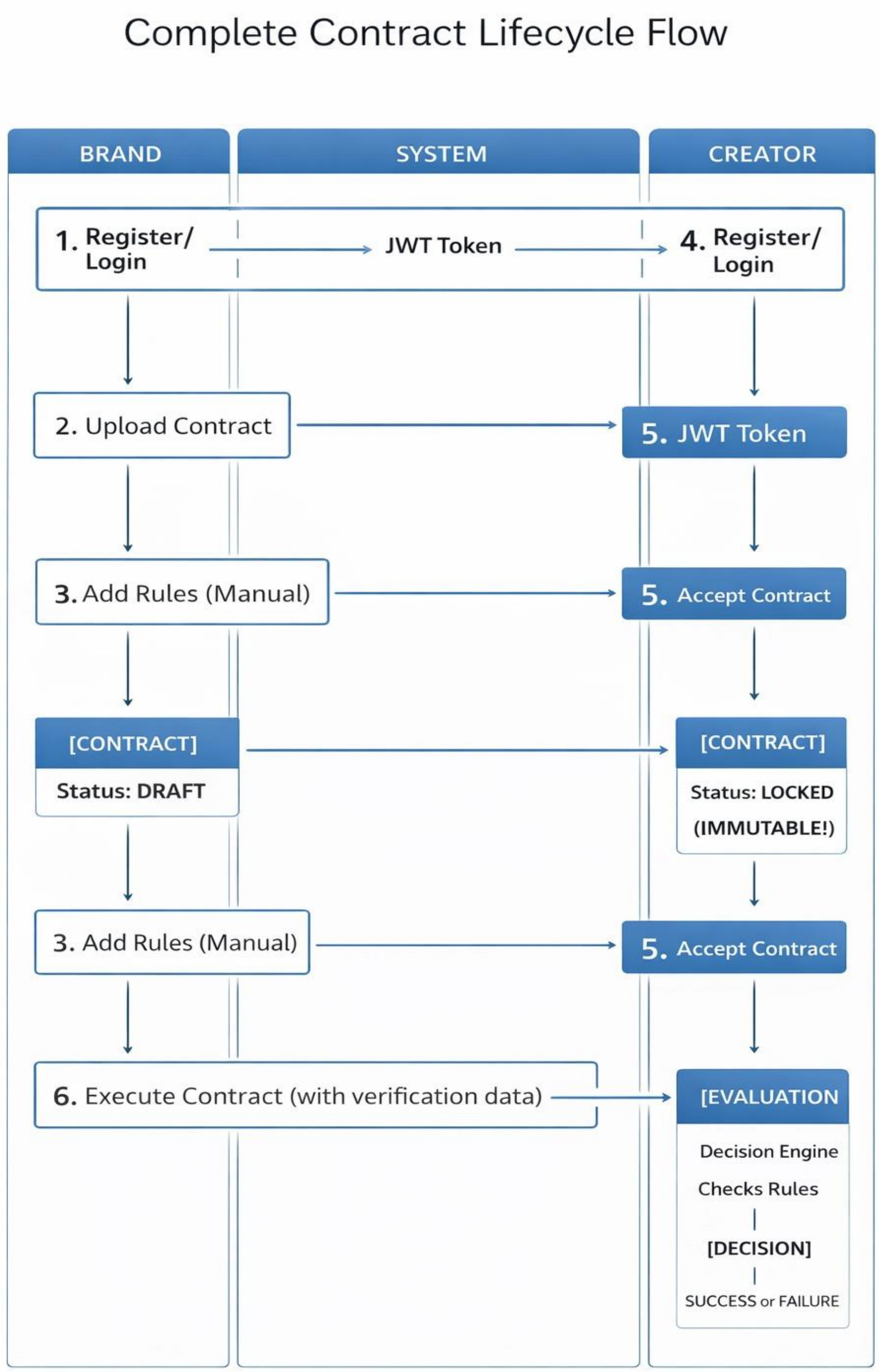

Figure 4.1.2.3 Complete Contract Lifecycle Flow

This diagram shows how a contract moves between a brand, a system, and a creator. Both log in, then the brand uploads a contract and adds rules. The creator accepts it, locking the contract so it can’t be changed. Later, the brand executes the contract, and the system evaluates whether the rules were followed. The outcome is either success or failure, ensuring fairness and immutability

4.1.3 Sequence Diagram : -

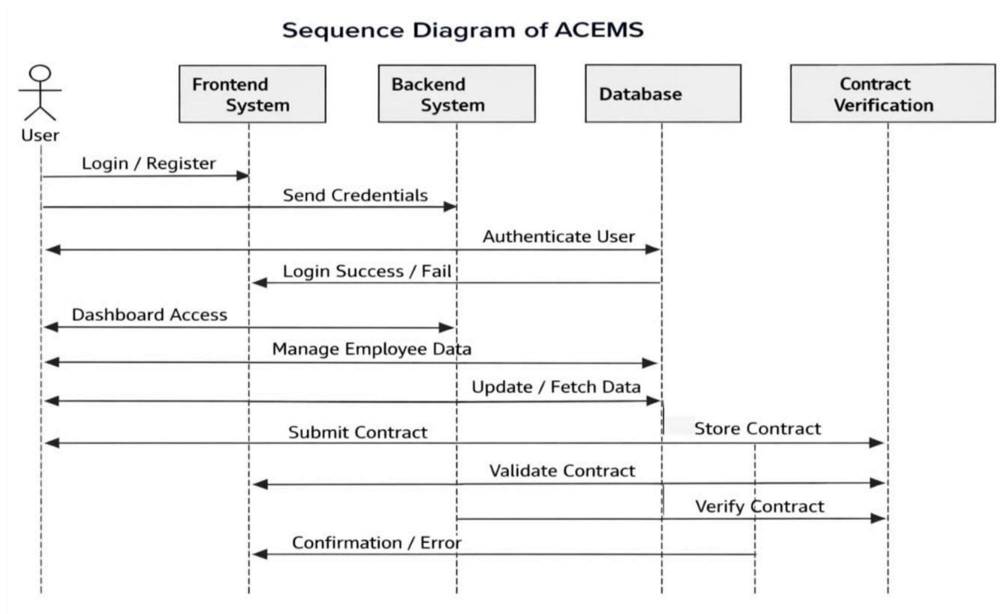

Figure 4.1.3.1 Sequence Diagram of ACEMS

The diagram illustrates how the ACEMS system manages user interactions, authentication, data handling, and contract verification. It shows the step-by-step flow of requests between the user, frontend, backend, database, and contract verification module, ensuring secure login, efficient employee data management, and reliable contract validation.

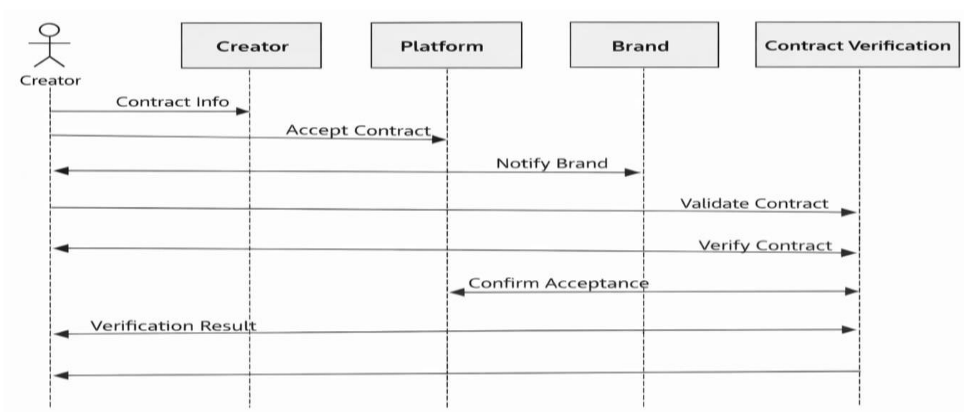

Figure 4.1.3.2 Creator Sequence Diagram

This sequence diagram outlines the contract acceptance and verification process between a creator, platform, brand, and contract verification system. It shows how the creator initiates contract acceptance, the platform informs the brand, and the brand ensures validation through the verification system before confirming back to the creator.

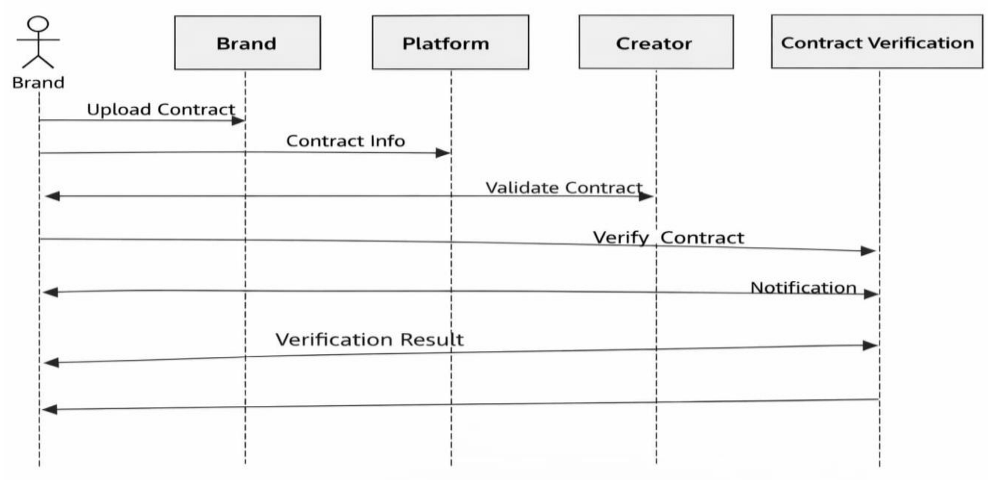

Figure 4.1.3.3 Brand Sequence Diagram

This diagram demonstrates the contract verification workflow among the brand, platform, creator, and verification system. It begins with the brand uploading a contract, which is passed to the platform and then to the creator for validation. The creator engages the contract verification system, receives a notification, and relays the verification result back through the platform to the brand, ensuring transparency and structured communication at each stage.

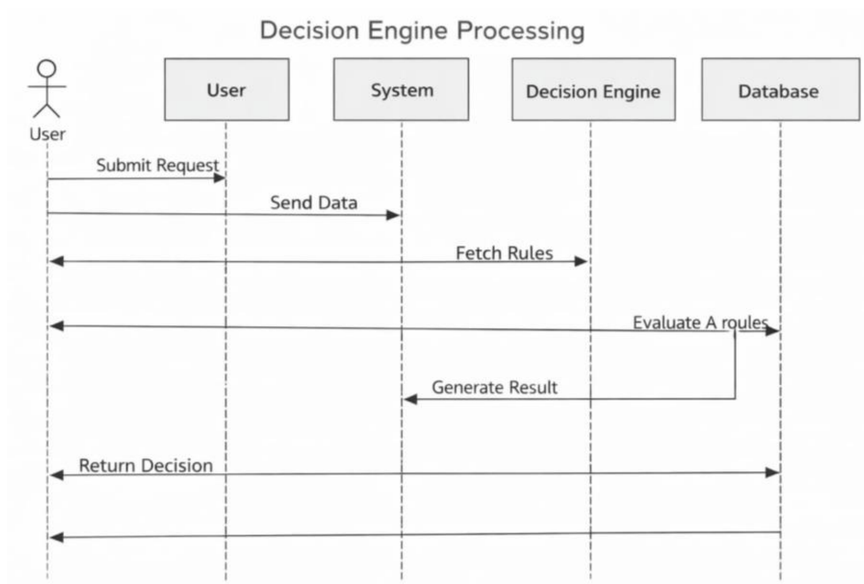

Figure 4.1. 3.4 Decision Engine Processing Diagram

This diagram highlights the workflow of a decision engine within a system. The user submits a request, which the system forwards to the decision engine. The engine retrieves rules from the database, evaluates them, and generates a result. Finally, the system returns the decision to the user, ensuring a structured and rule-based process for handling requests.

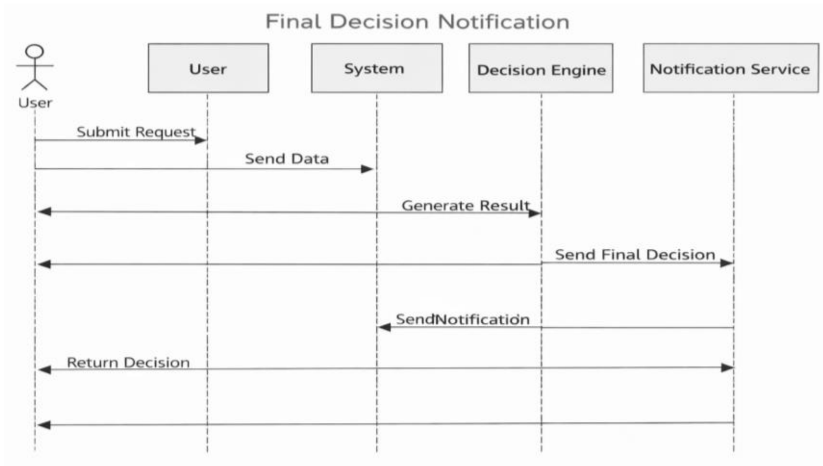

Figure 4.1.3.5 Final Decision Notification System

This diagram illustrates the notification flow for final decisions in a system. The user submits a request, which the system processes through the decision engine. Once the engine generates a result, it is passed to the notification service, which relays the outcome back through the system to the user—ensuring that decisions are both processed and communicated in a structured, reliable manner.

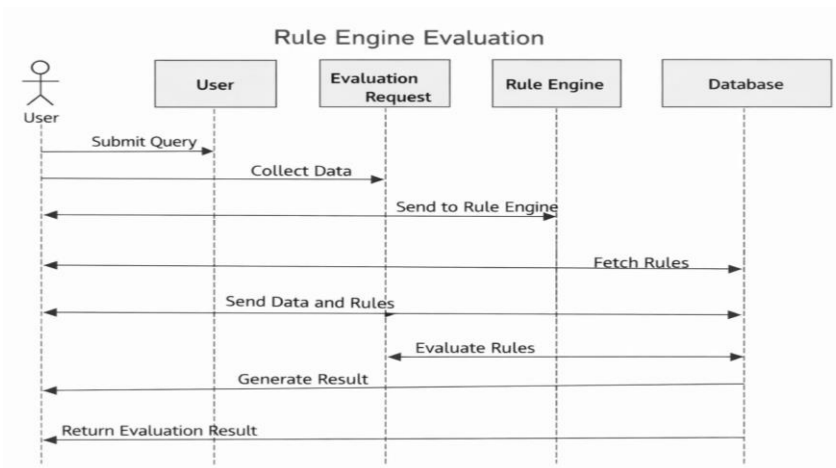

Figure 4.1.3.6 Rule Engine Evaluation

This diagram explains the Rule Engine Evaluation workflow. The user submits a query, which is collected as an evaluation request and sent to the rule engine. The rule engine fetches rules from the database, evaluates them against the provided data, and returns the results through the evaluation request back to the user. This structured flow ensures that queries are processed consistently and decisions are generated based on predefined rules.

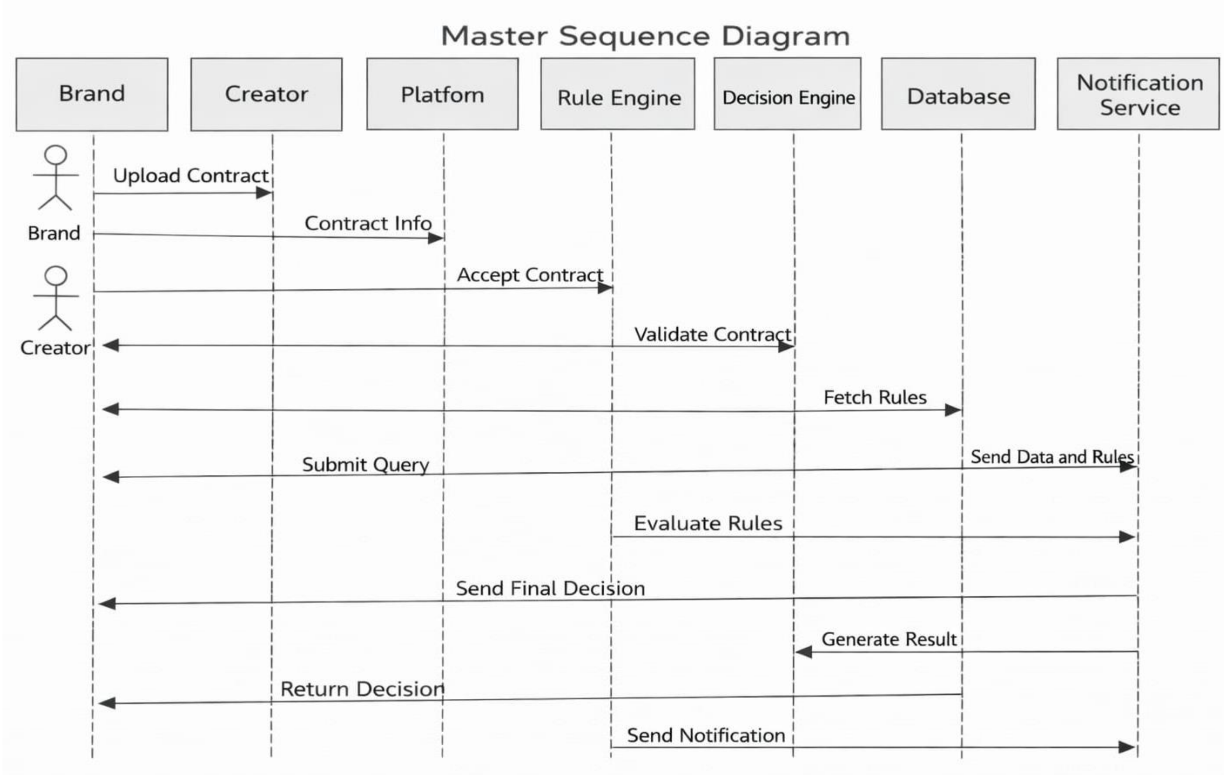

Figure 4.1.3.7 Master Sequence Diagram

This Master Sequence Diagram shows the complete workflow across multiple system components—brand, creator, platform, rule engine, decision engine, database, and notification service. It begins with contract upload and acceptance, moves through rule evaluation and decision-making, and ends with notifications being sent. By mapping each interaction step-by-step, it highlights how contracts, queries, and rules are processed in a structured flow to ensure accurate validation and timely communication.

CHAPER 7 CONCLUSION

Autonomous Contract Execution and Monitoring System (ACEMS) is one of the effective solutions to addressing the problem of digital contracts between content creators and brands. Conventionally managed contract processes are prone to manual monitoring that might be time consuming, conflicts and lack of transparency. The proposed system automatizes the contract implementation and ensures constant monitoring of the contract events to ensure that all the set terms and deliverables of the contract are achieved.

The system increase reliability, accountability, and transparency in the influencer marketing partnerships by establishing automated checks and balances systems. It reduces the use of hands and helps in maintaining an orderly record of the activities in the contract at the various stages of the contract lifecycle.

Overall, ACEMS demonstrates that digital contract management systems can be enhanced with the help of automation and intelligent monitoring since the collusion between stakeholders will be more efficient, secure and trustworthy.

CHAPTER 8 REFERENCE

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